



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

21-08-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	B	C	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C,D	A,B,D	A,B,D	A,B,C,D	A,B,D	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.33	0.33	85.33	96.00	2.38 to 2.39	0.50
SECTION-II (ii)	Q.	7	8	9			
	A.	8	7	3			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	C	B	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C	A,B,C	A,B,D	A,C,D	A,B,D	A,B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	6	7	6.30	10.30	7.00	6.64 or 6.65
SECTION-II (ii)	Q.	7	8	9			
	A.	8	6	3			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	B	C	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	B,C,D	A,B,D	A,B,D	A,B,C	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	10.00	45.00	0.10	1.00	1.33	3.00
SECTION-II (ii)	Q.	7	8	9			
	A.	7	9	4			

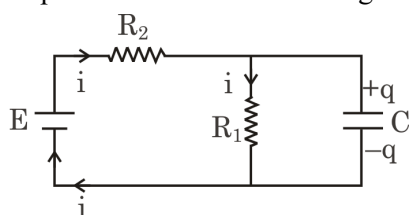
HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

Capacitor has maximum charge at steady state



$$\frac{q}{C} = iR_1$$

$$q = \frac{CER_1}{R_1 + R_2}$$

$$\ln q = \ln C + \ln E + \ln R_1 - \ln(R_1 + R_2)$$

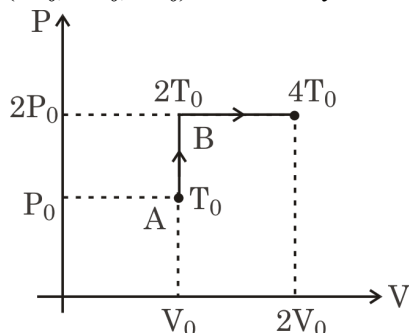
$$\frac{\Delta q}{q} = \frac{\Delta C}{C} + \frac{\Delta E}{E} + \frac{\Delta R_1}{R_1} - \left(\frac{\Delta R_1 + \Delta R_2}{R_1 + R_2} \right)$$

$$\frac{\Delta q}{q} = \frac{\Delta C}{C} + \frac{\Delta E}{E} + \Delta R_1 \left\{ \frac{1}{R_1} - \frac{1}{R_1 + R_2} \right\} - \frac{\Delta R_2}{R_1 + R_2}$$

$$\left| \frac{\Delta q}{q} \right|_{\max} = \frac{0.01}{1} + \frac{4}{40} + 3 \left\{ \frac{1}{60} - \frac{1}{90} \right\} + \frac{6}{90} = 0.19$$

2. Ans (B)

Let initial pressure, volume, temperature be P_0 , V_0 , T_0 indicated by state A in P-V diagram. The gas is then isochorically taken to state B ($2P_0$, V_0 , $2T_0$) and then taken from state B to state C ($2P_0$, $2V_0$, $4T_0$) isobarically.



Total heat absorbed by 1 mole of gas

$$\Delta Q = C_v (2T_0 - T_0) + C_p (4T_0 - 2T_0)$$

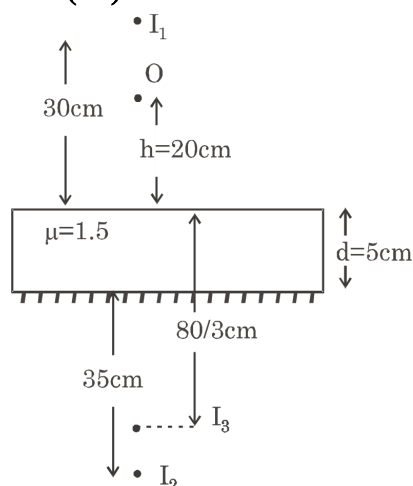
$$= \frac{5}{2}RT_0 + \frac{7}{2}R \times 2T_0 = \frac{19}{2}RT_0$$

Total change in temperature from state A to C is $\Delta T = 3T_0$

$\therefore$ Molar heat capacity

$$= \frac{\Delta Q}{\Delta T} = \frac{\frac{19}{2}RT_0}{3T_0} = \frac{19}{6}R$$

3. Ans (C)



Air $\rightarrow$ Glass Refraction :

$$\frac{1.5}{V} - \frac{1}{-20} = 0$$

$$V = -30\text{cm} (I_1)$$

For Mirror : Image is I_2

Glass $\rightarrow$ Air Refraction

$$\frac{1}{V} - \frac{1.5}{-40} = 0$$

$$V = \frac{-80}{3}\text{cm} (I_3)$$

4. Ans (B)

If the kinetic energy of each particle is k , then

$$2k + 2(0.51\text{MeV}) = \frac{hc}{\lambda}$$

$$= \frac{12.4 \times 10^{-3}\text{MeV}\text{Å}}{5 \times 10^{-4}\text{Å}}$$

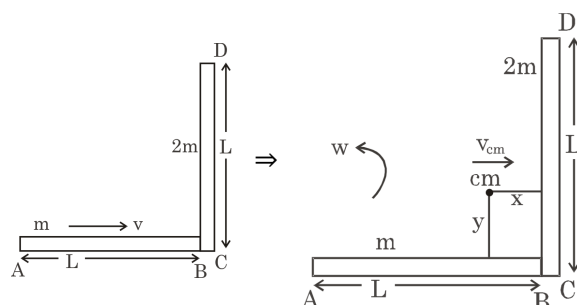
$$\Rightarrow k = \frac{24.8 - 1.022}{2} = 11.9\text{MeV}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,B,C,D)

Colm :



$$mv = 3m v_{cm} \quad K.E_t = \frac{mv^2}{6}$$

$$y = \frac{2m(L/2)}{3m}$$

$$v_{cm} = \frac{v}{3} \quad x = \frac{ML/2}{3m}$$

$$y = \frac{L}{3}$$

$$K.E_t = \frac{1}{2}(3m) v_{cm}^2 \quad x = \frac{L}{6}$$

$$I_{cm} = \frac{ML^2}{12} + M \left[\frac{L^2}{9} + \frac{L^2}{9} \right] + \frac{2mL^2}{12} + 2m \left[\frac{L^2}{36} + \frac{L^2}{36} \right]$$

$$I_{cm} = \frac{7}{12}ML^2$$

Coam :

$$mvy = I_{cm} \omega$$

$$\frac{mvL}{3} = \frac{7}{12}mL^2 \omega$$

$$\omega = \frac{4v}{7L}$$

$$K.E_i = \frac{1}{2}mv^2$$

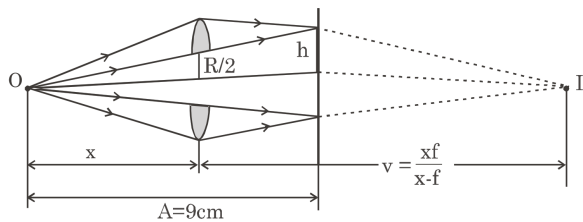
$$K.E_f = \frac{1}{2}(3m) v_{cm}^2 + \frac{1}{2}I_{cm} \omega^2$$

$$K.E_f = \frac{mv^2}{6} + \frac{1}{2} \left(\frac{7}{12}mL^2 \right) \left(\frac{4v}{7L} \right)^2$$

$$K.E_f = \frac{11}{42}mv^2$$

$$\text{loss in K.E.} = \frac{1}{2}mv^2 - \frac{11}{42}mv^2 = \frac{5}{21}mv^2$$

6. Ans (A,B,D)



$$\text{For lens : } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-x} = \frac{1}{f}$$

$$v = \frac{xf}{x-f} \quad \dots(i)$$

Similar triangle

$$\frac{h}{A} = \frac{R/2}{x} \quad \dots(ii)$$

$$\frac{R}{v} = \frac{h}{v - (A - x)} \quad \dots(iii)$$

from (i), (ii) & (iii)

$$2x^2 - 2Ax + Af = 0$$

$$x = \frac{A}{2} + \frac{1}{2} \sqrt{A(A-2f)}$$

For $A = 9\text{cm}$, $f = 4\text{cm} \Rightarrow x = 6\text{cm}$ & $x = 3\text{cm}$

$x = 6\text{cm} \rightarrow$ real image

$x = 3\text{cm} \rightarrow$ virtual image

Phenomena possible only for $A > 2f$

7. Ans (A,B,D)

The disc will stop (translation) at time $t = \frac{v_0}{\mu g}$

Since, $\tau = I\alpha$

$$\mu Mg \cdot R = \frac{MR^2}{2} \cdot \alpha$$

$$\alpha = \frac{2\mu g}{R}$$

and $\omega = \omega_0 - \alpha t$

$$\downarrow$$

$$0 = \frac{2v_0}{R} - \frac{2\mu R}{R} t \Rightarrow t = \frac{v_0}{\mu g}$$

Also at the same time, linear speed also ceases.

$$V = v_0 - \mu g t$$

$$0 = v_0 - \mu g t$$

$$T = v_0 / \mu g$$

Thus it will not regain ω .

Since linear speed and angular speed becomes zero at the same time.

8. Ans (A,B,C,D)

If energy required by transition or ionization is

Q then kinetic energy K required for colliding.

Particle is

$$K = Q \left(1 + \frac{M_1}{M_2} \right) \text{ here } M_1 = M_2 \therefore K = 2Q$$

Minimum kinetic energy required for neutron

to excite the atom = $2 \times$ excitation energy.

Possible values of excitation energy = 10.2 eV,

12.1 eV,.... Ionization energy = 13.6 eV.

9. Ans (A,B,D)

Resistance = 0

$\Phi_{\text{net}} = \text{constant}$

$$\Phi_i = \Phi_f$$

$$BA = BA \cos \theta + Li$$

$\theta =$ angular displacement

$i =$ current

$$i = \frac{BA}{L} (1 - \cos \theta)$$

$$i|_{\theta=\frac{\pi}{3}} = \frac{BA}{2L} = \frac{Blb}{2L}$$

$$i_{\text{max}} = \frac{2BA}{L} = \frac{2Blb}{L} \text{ (at } \theta = \pi \text{)}$$

$$\tau = MB \sin \theta$$

$$\tau = -iAB \sin \theta$$

$$\tau = \frac{-BA}{L} (1 - \cos \theta) AB \sin \theta$$

$$\tau \text{ is max when } \frac{d\tau}{d\theta} = 0$$

$$\cos \theta - \cos 2\theta = 0$$

$$\theta = \frac{2\pi}{3}$$

$$|\tau|_{\text{max}} = \frac{3\sqrt{3} B^2 A^2}{4L}$$

10. Ans (A,B,C,D)

V_x = volume of block inside lower liquid

Force balance :

$$\rho \cdot V g = \rho_1 g V_x + \rho_2 g (V - V_x)$$

$$V_x = \frac{V (\rho - \rho_2)}{\rho_1 - \rho_2}$$

The level of the bottom liquid rises by the V_x .
Because the area of the vessel's base is S then
the level of the interface between the liquids
rises by Δh ,

$$\Delta h = \frac{V_x}{S}$$

$$\text{Replacing } V_x \text{ here we get } \Delta h = \frac{V (\rho - \rho_2)}{S (\rho_1 - \rho_2)}$$

$$\text{Level of top liquid changes by } \frac{V}{S}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (0.33)

$$v = \sqrt{2ag} \sin \theta$$

$$\sin \theta = \frac{v}{\sqrt{2ag}}$$

$$\cos \theta = \sqrt{\frac{1 - v^2}{2ag}}$$

position (x,y) at

$$t = \frac{2a}{v}$$

$$x = \sqrt{2ag} \cos \theta \left(\frac{2a}{v} \right)$$

$$x = \sqrt{2ag} \cos \theta \left[\frac{2a}{\sqrt{2ag} \sin \theta} \right]$$

$$x = 2a \cot \theta \quad \dots(i)$$

$$y = \sqrt{2ag} \sin \theta \left(\frac{2a}{v} \right) - \frac{1}{2} g \left(\frac{2a}{v} \right)^2$$

$$y = \sqrt{2ag} \sin \theta \left[\frac{2a}{\sqrt{2ag} \sin \theta} \right] - \frac{2a^2 g}{2ag \sin^2 \theta}$$

$$y = 2a - a \sec^2 \theta$$

$$y = 2a - a (1 + \cot^2 \theta)$$

$$y = 2a - a \left[1 + \frac{x^2}{4a^2} \right]$$

$$y = a - \frac{x^2}{4a} \Rightarrow 4ay + x^2 = 4a^2$$

$$\Rightarrow C_1 = 4 \text{ \& } C_2 = 2$$

$$\frac{C_2}{C_1 + C_2} = 0.33$$

2. Ans (0.33)

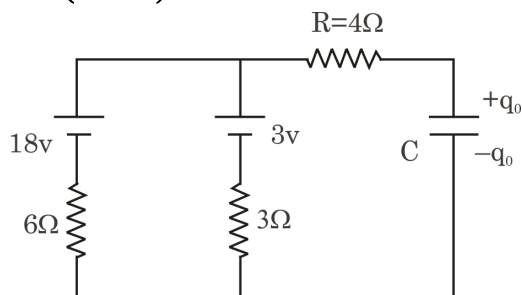
$$\vec{V}_f = \sqrt{2ag} \cos \theta \hat{i} + \left[\sqrt{2ag} \sin \theta - g \left(\frac{2a}{v} \right) \right] \hat{j}$$

$$\vec{V}_i = \sqrt{2ag} \cos \theta \hat{i} + [\sqrt{2ag} \sin \theta] \hat{j}$$

$$\vec{V}_f \cdot \vec{V}_i = 2ag \cos^2 \theta + 2ag \sin^2 \theta - \sqrt{2ag} \sin \theta g \left[\frac{2a}{\sqrt{2ag} \sin \theta} \right]$$

$$\vec{V}_f \cdot \vec{V}_i = 0 \Rightarrow \vec{V}_f \perp \vec{V}_i \Rightarrow K = \frac{1}{3}$$

3. Ans (85.33)



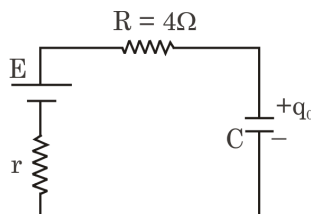
$$C = 16 \mu F$$

$$q_0 = 64 \mu C$$

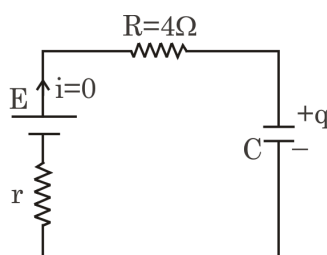
$$E = \frac{\frac{18}{6} + \frac{3}{3}}{\frac{1}{6} + \frac{1}{3}} = 8V$$

$$r = 6/3 = 2\Omega$$

at $t = 0$



At $t = \infty$



$$q = CE$$

$$q = 128 \mu C$$

$$W_b = H + \Delta U$$

$$(q - q_0) E = H + \frac{q^2}{(2C)} - \frac{q_0^2}{(2C)}$$

$$(128 - 64) 8 = H + \frac{128^2}{2 \times 16} - \frac{64^2}{2 \times 16}$$

$$H = 128 \mu J \text{ (Total heat loss)}$$

$$H' = \left(\frac{4}{2+4} \right) (128)$$

$$H' = 85.33 \mu J$$

4. **Ans (96.00)**

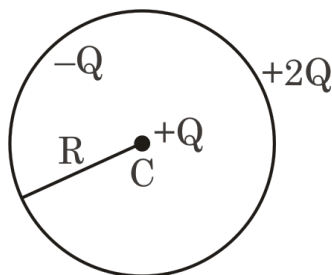
$$\tau = (R + r) C$$

$$\tau = (4 + 2) \times 16$$

$$\tau = 96 \mu s$$

6. **Ans (0.50)**

Charge - Q must be induced on inner surface.



So + 2Q charge appears on outer surface.

Charges on inner and outer surface distribute uniformly.

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (8)**

Radius of curvature of the curve

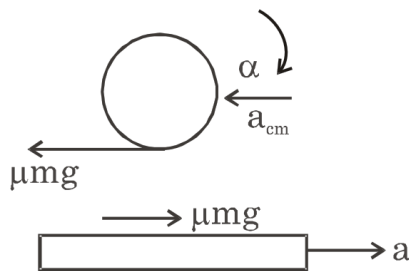
$$r = \frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}{\left| \frac{d^2y}{dx^2} \right|}$$

$$y = 2x^2, \frac{dy}{dx} = 4x, \frac{d^2y}{dx^2} = 4$$

$$r = \frac{[1 + 16x^2]^{3/2}}{4} \Rightarrow \text{at } x = 0, r = \frac{1}{4}$$

$$r = \frac{mv}{qB} \Rightarrow B = \frac{mV}{qr} = 8$$

8. **Ans (7)**



For cylinder : $\mu mg = ma_{cm}$

$$a_{cm} = \mu g \leftarrow$$

$$\tau_{cm} = I_{cm} \alpha$$

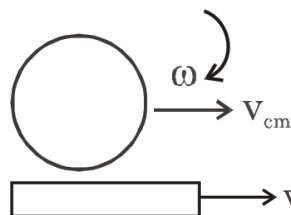
$$\mu mg R = \frac{mR^2 \alpha}{2}$$

$$\alpha = \frac{2\mu g}{R}$$

For plank $\mu mg = 2ma$

$$a = \frac{\mu g}{2} \rightarrow$$

At time t



When slipping ends : $v_{cm} - R\omega = v$

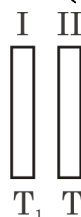
$$(v_0 - a_{cm}t) - R(\alpha t) = at$$

$$v_0 = \frac{7}{2} \mu gt$$

$$t = \frac{2v_0}{7\mu g}$$

$$x = \frac{2}{7}$$

9. **Ans (3)**



$P = 2\sigma AT_1^4 - \sigma AT_2^4$ (steady state condition of sheet I) (i)

and $2\sigma AT_2^4 = \sigma AT_1^4$ (steady state condition of sheet II) (ii)

$$\Rightarrow P = 3\sigma AT_2^4 = 3kW$$

PART-2 : CHEMISTRY

SECTION-I (i)

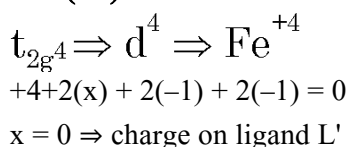
3. **Ans (B)**

$$= \left[3.14 \times \frac{8RT}{\pi M} + 2 \left(\frac{2RT}{M} - \frac{3RT}{M} \right) \right] \times 10^{-7}$$

$$= \frac{6RT \times 10^{-7}}{M}$$

$$= \frac{6 \times 8.314 \times 2000 \times 10^{-7}}{8.314 \times 4 \times 10^{-3}} \Rightarrow 0.3$$

4. **Ans (C)**



PART-2 : CHEMISTRY

SECTION-I (ii)

6. **Ans (A,B,C)**

Calculation of mass percentage of carbon and hydrogen :

Percentage of carbon

$$= \frac{12}{44} \times \frac{\text{mass of CO}_2}{\text{mass of compound}} \times 100$$

$$= \frac{12}{44} \times \frac{0.792}{0.450} \times 100 = 48$$

Percentage of hydrogen

$$= \frac{2}{18} \times \frac{0.324}{0.450} \times 100 = 8$$

Calculation of mass percentage of nitrogen :

$$\text{Amount of H}_2\text{SO}_4 \text{ taken} \equiv 25 \text{ mL of } \frac{N}{2} \text{ H}_2\text{SO}_4$$

$$\equiv 12.5 \text{ mL of 1N H}_2\text{SO}_4$$

$$\text{Amount of unreacted H}_2\text{SO}_4 \equiv 38.5 \text{ mL of } \frac{N}{5} \text{ NaOH solution}$$

$$\equiv 7.7 \text{ mL of 1N NaOH solution}$$

$$\equiv 7.7 \text{ mL of 1N H}_2\text{SO}_4 \text{ solution}$$

$$\therefore \text{amount of H}_2\text{SO}_4 \text{ that reacted with ammonia}$$

$$\equiv (12.5 - 7.7) \text{ mL of 1N H}_2\text{SO}_4$$

$$\equiv 4.8 \text{ mL of 1N H}_2\text{SO}_4$$

So, amount of ammonia $\equiv 4.8 \text{ mL of 1N}$ ammonia solution.

1000 mL of 1N ammonia solution $\equiv 17 \text{ g of}$ ammonia = 14 g of nitrogen.

$$\therefore 4.8 \text{ mL of 1N ammonia solution} \equiv \frac{14 \times 4.8}{1000} \text{ g}$$

of nitrogen

$$\equiv 0.067 \text{ g of nitrogen.}$$

So, 0.240 g of the compound contains 0.0672 g of nitrogen.

$$\therefore \text{percentage of nitrogen} = \frac{0.0672}{0.240} \times 100 = 28$$

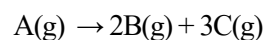
7. **Ans (A,B,D)**

(A) Colligative property μ number of particles of solute.

(C) $\Delta G_{\text{solution}} < 0$ for solution

$$(D) \Delta S_{\text{surround}} = -\frac{\Delta_r H}{T} = -\frac{(-ve)}{T} \Rightarrow +ve$$

8. **Ans (A,C,D)**



$$t=0 \quad P_A \quad 0 \quad 0$$

$$t=90 \text{ sec} \quad P_A - x \quad 2x \quad 3x$$

$$t=\infty \quad 0 \quad 2P_A \quad 3P_A$$

$$P_0 = P_A + P' = 500$$

$$P_t = P_A + 4x + P' = 1900$$

$$P_\infty = 5P_A + P' = 2100$$

$$P_A = 400, P' = 100, x = 350$$

$$(P_A)_t = \frac{P_A}{2^n}, n = 3, \quad 90 = 3t_{1/2}$$

$$t_{1/2} = 30 \text{ sec}$$

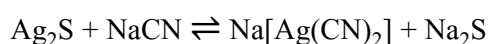
$$(A) t_{3/4} = 2t_{1/2} = 60 \text{ sec}$$

$$(B) P_A \text{ at 75\% conversion} = 400 \times \frac{25}{100} = 100 \text{ mm Hg}$$

$$(C) t_{\text{avg}} = \frac{1}{K} = \frac{t_{1/2}}{\ln 2} = \frac{30}{0.7} = 42.86 \text{ sec}$$

$$(D) P_{\text{total}} = 200 + 5 \times 200 + 100 \Rightarrow 1300 \text{ mm Hg}$$

9. **Ans (A,B,D)**



on addition of O_2 into solution, oxidation of

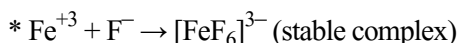
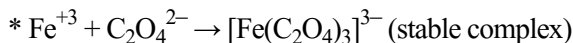
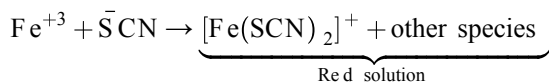
Na_2S takes place $\Rightarrow$ eqbm shifts forward

* Pt is less reactive, it can not reduce Ag.

10. **Ans (A,B,C)**

Yellow coloured solution of metal ion belonging to

basic radicals of group-III $\Rightarrow \text{Fe}^{+3}$ is metal ion



F^- and $\text{C}_2\text{O}_4^{3-}$ are stronger ligand than SCN^- .

Hg^{2+} forms stable complex with SCN^-

PART-2 : CHEMISTRY

SECTION-II (i)

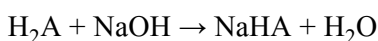
1. **Ans (6)**

$$x = 2, t = 4$$

2. **Ans (7)**

$$y = 6, p = 1$$

4. **Ans (10.30)**

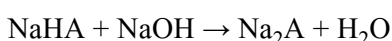


$$P = \frac{1}{2}(\text{pK}_{a_1} + \text{pK}_{a_2}) = \frac{1}{2}(4.6 + 8) = 6.3$$

$$N_1 V_1 = N_2 V_2$$

$$N_1 \times 10 = 0.1 \times 20$$

$$N_1 = 0.2$$



$$C = \frac{2m \text{ moles}}{50\text{ml}} \Rightarrow \frac{1}{25} \text{M}$$

$$Q = 7 + \frac{1}{2}(\text{pK}_{a_2} + \log C) = 7 + \frac{1}{2}(8 - 2 \times 0.7)$$

$$Q = 10.3$$

6. **Ans (6.64 or 6.65)**

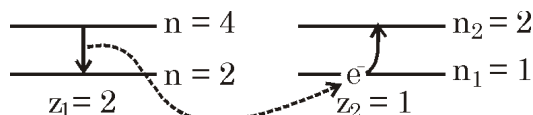
$$\text{Energy of emitted photon} = 13.6 \times (2)^2 \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = 10.2 \text{ eV}$$

$$e = \psi + \text{KE}_{\text{max}}$$

$$10.2 \text{ eV} = 3.2 \text{ eV} + \text{KE}_{\text{max}}$$

$$\text{KE}_{\text{max}} = 7 \text{ eV}$$

$$x = 7$$



$$10.2 = +13.6 \times 1 \left(\frac{1}{1^2} - \frac{1}{x^2} \right) \Rightarrow n = 2$$

$$2\pi r = n\lambda$$

$$\lambda = 2\pi \times 0.529 \times 2 \Rightarrow 6.64 \text{ \AA}$$

$$y = 6.64$$

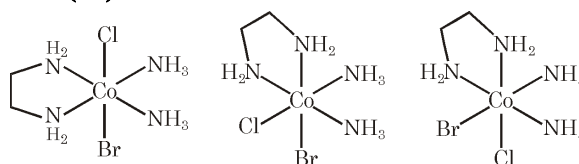
PART-2 : CHEMISTRY

SECTION-II (ii)

8. **Ans (6)**

Compound C is 4-methylpentan-2-one

9. **Ans (3)**



PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A)**

$$2^{40} = 2^4 \cdot 3^1 \cdot 5^1$$

$$1(1+1)(1+1) = 4$$

2. **Ans (B)**

$$\frac{\left(x + \frac{1}{x}\right)^4 - \left(x^4 + \frac{1}{x^4}\right) - 1}{\left(x + \frac{1}{x}\right)^2 + \left(x^2 + \frac{1}{x^2}\right)}$$

$$\text{Put } x + \frac{1}{x} = t$$

$$\Rightarrow y = \frac{t^4 - [(t^2 - 2)^2 - 2] - 1}{t^2 + t^2 - 2} = \frac{t^4 - [t^4 - 4t^2 + 2] - 1}{2(t^2 - 1)}$$

$$\Rightarrow y = \frac{4t^2 - 3}{2(t^2 - 1)} \Rightarrow y = \frac{4t^2 - 4 + 1}{2(t^2 - 1)} = \left(2 + \frac{1}{2(t^2 - 1)}\right)$$

$$\text{As } t = x + \frac{1}{x} \Rightarrow t^2 \geq 4 \Rightarrow t^2 - 1 \geq 3 \Rightarrow \frac{1}{t^2 - 1} \leq \frac{1}{3}$$

$$\Rightarrow \text{Maximum value of } y \text{ is } 2 + \frac{1}{3 \times 2} = \frac{13}{6}$$

3. Ans (C)

$$a_1 = 1$$

$$a_2 = a_1 + 4 = 1 + 4 = 5$$

$$a_3 = a_2 + 6 = 5 + 6 = 11$$

$$a_4 = a_3 + 8 = 11 + 8 = 19 \text{ and so on hence}$$

$$S = 1 + 5 + 11 + 19 + \dots + a_n$$

$$S = +1 + 5 + 11 + \dots + a_{n-1} + a_n$$

(-)

$$0 = 1 + 4 + 6 + 8 + \dots + (a_n - a_{n-1}) - a_n$$

$$a_n = 1 + 2(2 + 3 + 4 + \dots + (a_n - a_{n-1}))$$

(n-1) terms

$$a_n = 1 + 2 \cdot \frac{n-1}{2} [4 + (n-2)]$$

$$= 1 + (n-1)(n+2) = n^2 + n - 1$$

$$\text{sum} = \sum a_n = \sum n^2 + \sum n - \sum 1$$

$$= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} - n$$

$$\frac{n}{6} [(n+1)(2n+1) + 3(n+1) - 6]$$

$$= \frac{n}{6} [2n^2 + 3n + 1 + 3n + 3 - 6]$$

$$= \frac{n}{6} [2n^2 + 6n - 2] = \frac{n(n^2 + 3n - 1)}{3}$$

4. Ans (A)

$$\text{Let } I + f = (3 + \sqrt{5})^5 \text{ and } f' = (3 - \sqrt{5})^5$$

As before, we can prove that $f + f' = 1$.

Hence the integer just greater than the number

$$(3 + \sqrt{5})^5 \text{ is } I + f + f'$$

$$= (3 + \sqrt{5})^5 + (3 - \sqrt{5})^5$$

$$= 2 \left[{}^5C_0 3^5 + {}^5C_2 3^3 (\sqrt{5})^2 + {}^5C_4 3 (\sqrt{5})^4 \right]$$

$$= 2(243 + 1350 + 375) = 2(1968) = 3936$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (B,C)

$$f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$$

$$f'(x) = \frac{(x^2 + 2x + 4)(2x - 3) - (x^2 - 3x - 6)(2x + 2)}{(x^2 + 2x + 4)^2}$$

$$f'(x) = \frac{5x(x+4)}{(x^2 + 2x + 4)^2}$$

$$f'(x) : \frac{+}{-4} \quad \frac{-}{0} \quad \frac{+}{0}$$

$$f(-4) = \frac{11}{6}, f(0) = -\frac{3}{2}, \lim_{x \rightarrow \pm\infty} f(x) = 1$$

$$\text{Range} : \left[-\frac{3}{2}, \frac{11}{6} \right], \text{ clearly } f(x) \text{ is into}$$

$$\text{and } \lim_{x \rightarrow \infty} (f(x))^x \rightarrow 1^\infty$$

$$\lim_{x \rightarrow \infty} x(f(x) - 1) = e^{-5}$$

6. Ans (B,C,D)

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 - x + 1} - (ax + b)) = 0$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{(x^2 - x - 11) - (ax + b)^2}{\sqrt{x^2 - x + 1} + (ax + b)} \right) = 0$$

Here $a > 0$

$$\text{Coeff. of } x^2 \text{ in numerator} = 0 \quad \dots(i)$$

$$\text{Coeff. of } x \text{ in numerator} = 0 \quad \dots(ii)$$

$$\Rightarrow 1 - a^2 = 0 \Rightarrow a = 1$$

$$-1 - 2ab = 0 \Rightarrow 2b = -1$$

$$\Rightarrow a + 2b = 0$$

Now verify (C) and (D)

7. Ans (A,B,D)

(A)

$$f'_+(0) = \lim_{h \rightarrow 0} \frac{h^2 \sin \frac{1}{h} - 0}{h} = \lim_{h \rightarrow 0} h \sin \left(\frac{1}{h} \right) = 0$$

$$(B) \lim_{x \rightarrow 0^+} \frac{x^2 \sin \left(\frac{1}{x} \right)}{x^2} = \lim_{x \rightarrow 0^+} \sin \left(\frac{1}{x} \right) = \text{DNE}$$

(C)

$$\lim_{x \rightarrow 0^+} \frac{2x \sin \left(\frac{1}{x} \right) + x^2 \cos \left(\frac{1}{x} \right) \left(-\frac{1}{x^2} \right)}{2x} = \text{DNE}$$

8. **Ans (A,B,D)**

$$2x^2 - x = 2 - x^3$$

$$x^3 + 2x^2 - x - 2 = 0$$

$$(x+2)(x^2-1) = 0$$

$$x = -2, 1, -1$$

Points of intersection are $(-2, 10), (1, 1), (-1, 3)$

$$m_1 = \frac{dy}{dx} = 4x - 1$$

$$m_2 = \frac{dy}{dx} = -3x^2$$

$$\text{at } (-2, 10) \Rightarrow m_1 = -9, m_2 = -12$$

$$\tan \theta = \frac{3}{109}$$

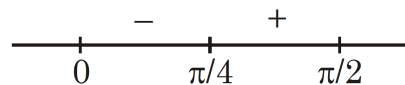
$$\text{at } (-1, 3) \Rightarrow m_1 = -5, m_2 = 3$$

$$\tan \theta = \frac{1}{8}$$

9. **Ans (A,B,C)**

$$f(x) = \sin^5 x + \cos^5 x - 1, x \in \left[0, \frac{\pi}{2}\right]$$

$$f'(x) = 5\sin^4 x \cos x - \cos^4 x \sin x = 5\sin x \cos x [\sin^3 x - \cos^3 x]$$



$$\text{also } f(0) = f\left(\frac{\pi}{2}\right) = 0 \Rightarrow (A) \text{ is correct}$$

$$f'\left(\frac{\pi}{4}\right) = 0 \Rightarrow (B) \text{ is also correct}$$

in $\left[0, \frac{\pi}{4}\right]$ according to Rolle's theorem $\Rightarrow f''(c_1) = 0$

also in $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ according to Rolle's theorem $\Rightarrow f''(c_2) = 0 \Rightarrow (C) \text{ is also correct}$

10. **Ans (A,B,C,D)**

$$(A) \vec{OX} = \vec{a} + \frac{3}{4}\vec{AB}$$

$$= \vec{a} + \frac{3}{4}(\vec{b} - \vec{a}) = \frac{\vec{a}}{4} + \frac{3}{4}\vec{b}$$

$$(B) \vec{AC} = 2\vec{b} - \vec{a}$$

$$(C) \vec{AY} = -\vec{a} + h\left(\frac{\vec{a}}{4} + \frac{3}{4}\vec{b}\right)$$

$$(D) -\vec{a} + h\left(\frac{\vec{a}}{4} + \frac{3}{4}\vec{b}\right) = m(2\vec{b} - \vec{a})$$

$$\Rightarrow -1 + \frac{h}{4} = -m \Rightarrow \frac{3h}{4} = 2m$$

$$h = \frac{8}{5}, m = \frac{3}{5}$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (10.00)**

$$\text{Let } x_2 = x_1 + y_1, y_1 \geq 1$$

[variables y_1, y_2 have been chosen since $x_3 = x_2$

$+ y_2, y_2 \geq 1, x_1, x_2, x_3$ are given to be unequal]

Then the given equation reduces to

$$x_1 + (x_1 + y_1) + (x_1 + y_1 + y_2) = 14$$

$$\text{i.e. } 3x_1 + 2y_1 + y_2 = 14, \text{ where } x_1, y_1, y_2 \geq 1$$

Hence the required number of integral solutions

$$= \text{coeff. of } x^{14} \text{ in } (x^3 + x^6 + x^9 + \dots) \times (x^2 + x^4 + x^6 + \dots) \times (x + x^2 + x^3 + \dots)$$

$$= \text{coeff. of } x^{14-6} \text{ in } (1 + x^3 + x^6 + \dots) \times (1 + x^2 + x^4 + \dots) \times (1 + x + x^2 + x^3 + \dots)$$

$$= \text{coeff. of } x^8 \text{ in } (1 + x^2 + x^4 + x^6 + x^8 + \dots + x^3 + x^5 + x^7 + x^9 + \dots + x^6 + x^8 + x^{10} + \dots)$$

$$\times (1 + x + x^2 + x^3 + x^4 + x^5 + x^6 + x^8 + \dots)$$

$$= \text{coeff. of } x^8 \text{ in } (1 + x^2 + x^3 + x^4 + x^5 + x^6 + x^7 + x^8 + \dots) \times (1 + x + x^2 + x^3 + x^4 + x^5 + x^6 + x^7 + x^8 + \dots)$$

$$= (1 + 1 + 1 + 1 + 1 + 2 + 1 + 2) = 10$$

2. **Ans (45.00)**

Let

$$x_1 = x + 1$$

$$x_2 = y + 2$$

$$x_3 = z + 3$$

where x, y, z are non negative integers $x + y + z = 8$

No of non negative integral solution $8^{+2}C_2 = {}^{10}C_2$

3. Ans (0.10)

$$f(x) + \sin x \cdot f(x + \pi) = \sin^2 x \quad \dots(1)$$

$$x \rightarrow x + \pi$$

$$f(x + \pi) + \sin(\pi + x) \cdot f(x + 2\pi) = \sin^2(\pi + x)$$

$$f(x + \pi) - \sin x f(x) = \sin^2 x \quad \dots(2)$$

$$\text{From (1), } -f(x + \pi) = \frac{\sin^2 x - f(x)}{\sin x}$$

$$\text{From (2), } f(x + \pi) = \sin^2 x + \sin x \cdot f(x)$$

$$\therefore \sin^2 x + \sin x \cdot f(x) = \frac{\sin^2 x - f(x)}{\sin x}$$

$$\sin^3 x + \sin^2 x \cdot f(x) = \sin^2 x - f(x)$$

$$f(x) [1 + \sin^2 x] = \sin^2 x (1 - \sin x)$$

$$\therefore f(x) = \frac{\sin^2 x (1 - \sin x)}{1 + \sin^2 x}$$

Now,

$$\int \frac{\sin^2 x (1 - \sin x)}{1 + \sin^2 x} dx = \int \frac{(1 + \sin^2 x) - (1 - \sin x)}{1 + \sin^2 x} dx$$

$$= x - \int \frac{(1 - \sin x) + \sin x + \sin^3 x}{1 + \sin^2 x} dx$$

$$= x - \int \frac{1}{1 + \sin^2 x} dx + \int \frac{\sin x}{2 - \cos^2 x} dx + \int \sin x dx$$

$$I_1 = \int \frac{\sec^2 x}{1 + \tan^2 x + \tan^2 x} dx = \frac{1}{2} \int \frac{\sec^2 x}{\tan^2 x + \frac{1}{2}} dx$$

Put $\tan x = t$

$$= \frac{1}{2} \int \frac{dt}{t^2 + \frac{1}{2}} = \frac{1}{2} \cdot \sqrt{2} \cdot \tan^{-1}(\sqrt{2}t)$$

$$I_1 = \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2} \cdot \tan x) = I_2 = \int \frac{\sin x}{2 - \cos^2 x} dx$$

Put $\cos x = t$

$$= - \int \frac{dt}{2 - t^2} = - \frac{1}{2\sqrt{2}} \ln \frac{\sqrt{2} + t}{\sqrt{2} - t}$$

$$= - \frac{1}{2\sqrt{2}} \ln \frac{\sqrt{2} + \cos x}{\sqrt{2} - \cos x}$$

$$\therefore I = x + \cos x - \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2} \tan x) + \frac{1}{2\sqrt{2}} \ln \frac{\sqrt{2} + \cos x}{\sqrt{2} - \cos x}$$

4. Ans (1.00)

$$f(x) + \sin x \cdot f(x + \pi) = \sin^2 x \quad \dots(1)$$

$$x \rightarrow x + \pi$$

$$f(x + \pi) + \sin(\pi + x) \cdot f(x + 2\pi) = \sin^2(\pi + x)$$

$$f(x + \pi) - \sin x f(x) = \sin^2 x \quad \dots(2)$$

$$\text{From (1), } -f(x + \pi) = \frac{\sin^2 x - f(x)}{\sin x}$$

$$\text{From (2), } f(x + \pi) = \sin^2 x + \sin x \cdot f(x)$$

$$\therefore \sin^2 x + \sin x \cdot f(x) = \frac{\sin^2 x - f(x)}{\sin x}$$

$$\sin^3 x + \sin^2 x \cdot f(x) = \sin^2 x - f(x)$$

$$f(x) [1 + \sin^2 x] = \sin^2 x (1 - \sin x)$$

$$\therefore f(x) = \frac{\sin^2 x (1 - \sin x)}{1 + \sin^2 x}$$

Now,

$$\int \frac{\sin^2 x (1 - \sin x)}{1 + \sin^2 x} dx = \int \frac{(1 + \sin^2 x) - (1 - \sin x)}{1 + \sin^2 x} dx$$

$$= x - \int \frac{(1 - \sin x) + \sin x + \sin^3 x}{1 + \sin^2 x} dx$$

$$= x - \int \frac{1}{1 + \sin^2 x} dx + \int \frac{\sin x}{2 - \cos^2 x} dx + \int \sin x dx$$

$$I_1 = \int \frac{\sec^2 x}{1 + \tan^2 x + \tan^2 x} dx = \frac{1}{2} \int \frac{\sec^2 x}{\tan^2 x + \frac{1}{2}} dx$$

Put $\tan x = t$

$$= \frac{1}{2} \int \frac{dt}{t^2 + \frac{1}{2}} = \frac{1}{2} \cdot \sqrt{2} \cdot \tan^{-1}(\sqrt{2}t)$$

$$I_1 = \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2} \cdot \tan x) = I_2 = \int \frac{\sin x}{2 - \cos^2 x} dx$$

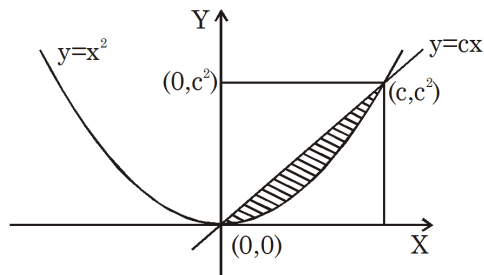
Put $\cos x = t$

$$= - \int \frac{dt}{2 - t^2} = - \frac{1}{2\sqrt{2}} \ln \frac{\sqrt{2} + t}{\sqrt{2} - t}$$

$$= - \frac{1}{2\sqrt{2}} \ln \frac{\sqrt{2} + \cos x}{\sqrt{2} - \cos x}$$

$$\therefore I = x + \cos x - \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2} \tan x) + \frac{1}{2\sqrt{2}} \ln \frac{\sqrt{2} + \cos x}{\sqrt{2} - \cos x}$$

5. Ans (1.33)

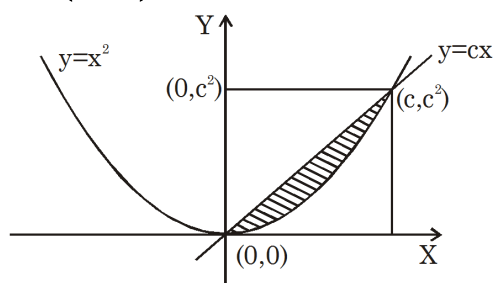


$$\text{Area}(T) = \frac{c \cdot c^2}{2} = \frac{c^3}{2}$$

$$\text{Area}(R) = \frac{c^3}{2} - \int_0^c x^2 dx = \frac{c^3}{2} - \frac{c^3}{3} = \frac{c^3}{6}$$

$$\therefore \lim_{c \rightarrow 0^+} \frac{\text{Area}(T)}{\text{Area}(R)} = \lim_{c \rightarrow 0^+} \frac{c^3}{2} \cdot \frac{6}{c^3} = 3$$

6. Ans (3.00)



$$\text{Area}(T) = \frac{c \cdot c^2}{2} = \frac{c^3}{2}$$

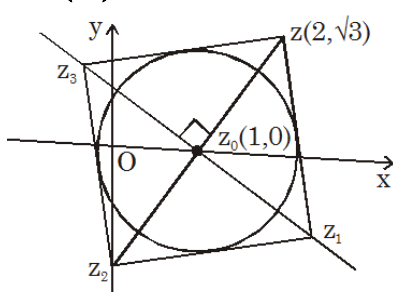
$$\text{Area}(R) = \frac{c^3}{2} - \int_0^c x^2 dx = \frac{c^3}{2} - \frac{c^3}{3} = \frac{c^3}{6}$$

$$\therefore \lim_{c \rightarrow 0^+} \frac{\text{Area}(T)}{\text{Area}(R)} = \lim_{c \rightarrow 0^+} \frac{c^3}{2} \cdot \frac{6}{c^3} = 3$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (7)



Since centre of circle i.e. (1, 0) is also the mid-point of diagonals of square

$$\Rightarrow \frac{z_1 + z_2}{2} = z_0 \Rightarrow z_2 = -\sqrt{3}i \text{ and } \frac{z_3 - 1}{z - 1} = e^{\pm i\pi/2}$$

$\Rightarrow$ other vertices are $z_3, z_1 = (1 - \sqrt{3}) + i$ and $(1 + \sqrt{3}) - i$.

8. Ans (9)

Let (l, m, n) be the direction ratios of the normal to the required plane so that $l - n = 0$ and $-1 + m = 0$

$\Rightarrow l = m = n$ and hence the equation of the plane containing (1, 1, 1) is $\frac{x}{3} + \frac{y}{3} + \frac{z}{3} = 1$.

Its intercepts with the coordinates axes are A(3, 0, 0); B(0, 3, 0); C(0, 0, 3).

Hence the volume of OABC

$$= \frac{1}{6} \begin{vmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{vmatrix} = \frac{27}{6} = \frac{9}{2} \text{ cubic units.}$$

9. Ans (4)

Any point on the parabola is $P(1 + t^2, 1 + 2t)$. The

equation of the tangent at P is

$t(y - 1) = x - 1 + t^2$ which meets the directrix $x = 0$ at

$Q\left(0, 1 + t - \frac{1}{t}\right)$. Let R be (h, k)

Since it divides QP externally in the ratio $\frac{1}{2} : 1$, Q

is the mid point of RP

$$\Rightarrow 0 = \frac{h + 1 + t^2}{2} \text{ or } t^2 = -(h + 1) \text{ and}$$

$$1 + t - \frac{1}{t} = \frac{k + 1 + 2t}{2} \text{ or } t = \frac{2}{1 - k}$$

$$\text{So that } \frac{4}{(1 - k)^2} + (h + 1) = 0 \text{ or } (k - 1)^2(h + 1)$$

$$+ 4 = 0$$

$$\text{Hence locus is } (y - 1)^2(x + 1) + 4 = 0$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

21-08-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C,D	A,C	A,B,C	A,B	A,C,D	B,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	B	B	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.00	3.00	40.00	60.00	140.00	0.50
SECTION-II (ii)	Q.	7	8	9			
	A.	4	9	3			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B	A,C	A,B,D	A,B,C	A,C,D	A,B,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	C	A	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	643.32	190.50	31.75	109.00	70.4	29.6
SECTION-II (ii)	Q.	7	8	9			
	A.	608	2	1			

PART-3 : MATHEMATICS

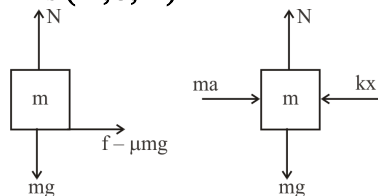
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C,D	A,C	A,B,D	A,B,C	A,B	A,B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	D	C	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	10.00	5.00	16.00	6.00	6.00	3.00
SECTION-II (ii)	Q.	7	8	9			
	A.	16	1	3			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B,C,D)



$$40 = 25 \times 2 - \frac{1}{2}a(2)^2 \Rightarrow a = 5 \text{ m/s}^2$$

$$\mu g = 5 \Rightarrow \mu = 0.5$$

$$kx = ma$$

$$500x = 10 \times 5 \Rightarrow x = 10 \text{ cm}$$

∴ Maximum elongation is = 20 cm

2. Ans (A,C)

Total energy released from $\text{Au}^{198} \rightarrow \text{Hg}^{198}$ in ground state $= (\Delta m_{\text{loss}})c^2$.

$$= (197.9682 - 197.9662) (930) = 1.86 \text{ MeV}$$

Energy released from ^{198}Hg in first excited state $\rightarrow \text{Hg}$ in ground state

$$= (-1.6) - (-2) \text{ MeV}$$

$\Rightarrow$ Energy released from $\text{Ag}^{198} \rightarrow \text{Hg}^{198}$ second excited state

$$= 1.86 - 0.4 = 1.46 = \text{max. K.E of } \beta_2 \text{ particle}$$

Similarly maximum kinetic energy of β_1 particle = 1.86 MeV

3. Ans (A,B,C)

$$\therefore V_S^2 = (3)^2 + (8 - 4)^2 = 25 \Rightarrow V_S = 5\text{volt}$$

$$Z = \frac{V_S}{I} = 5\Omega$$

$$V_R = 3 = 1 \times R \Rightarrow R = 3\Omega$$

$$P.F. = \frac{R}{Z} = \frac{3}{5} = 0.6$$

$$\therefore V_L > V_C \Rightarrow I \text{ lags } V$$

4. Ans (A,B)

Given refractive index $\mu = \sqrt{3}$

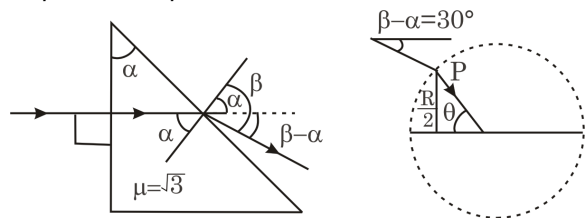
Angle of prism $\alpha = 30^\circ$

According to Snell's law.

$$\mu \sin \alpha = 1 \cdot \sin \beta$$

$$\sqrt{3} \sin 30^\circ = \sin \beta$$

$$\Rightarrow \beta = 60^\circ \Rightarrow \beta - \alpha = 30^\circ$$



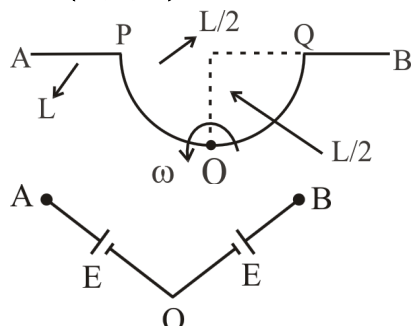
$$\sin \theta = \frac{\left(\frac{R}{2}\right)}{R} = \frac{1}{2} \Rightarrow \theta = 30^\circ = \beta - \alpha$$

$\therefore$ The incident ray on the sphere is directed radially.

$\therefore$ It will be reflected back and will retrace the path.

$\therefore$ Angle of deviation is 180° .

5. Ans (A,C,D)



$$V_A - V_0 = E; V_B - V_0 = E$$

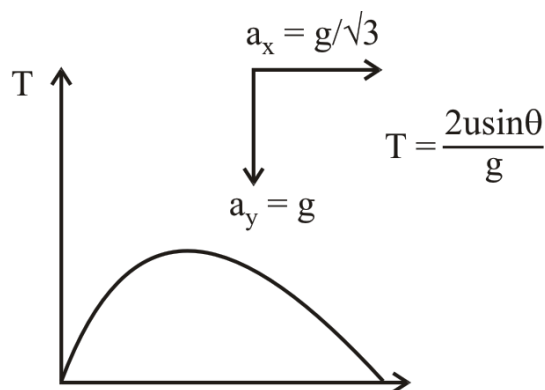
$$V_A - V_B = 0$$

$$(A) V_Q - V_0 = \frac{1}{2} \omega B \left(\frac{L}{\sqrt{2}} \right)^2$$

$$(C) V_Q - V_0 = \frac{1}{4} \omega B L^2$$

$$(D) V_Q - V_0 = +ve; V_P - V_0 = +ve$$

6. Ans (B,D)



$$R = u \cos \theta \frac{2u \sin \theta}{g} + \frac{1}{2} \frac{g}{\sqrt{3}} \left(\frac{2u \sin \theta}{g} \right)^2$$

$$= \frac{2u^2}{g} \left[\cos \theta \sin \theta + \frac{1}{\sqrt{3}} \sin^2 \theta \right]$$

$$\frac{dR}{d\theta} = 0, \tan 2\theta = -\sqrt{3}$$

$$\theta = 60^\circ$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (C)

$$P_0 + 0.8 \rho_w g h + \rho_w g (2h) + 1.2 \rho_w g (3h) = P_0 + 1.2 \rho_w g h_2$$

$$\Rightarrow 1.2 h_2 = h [0.8 + 2 + 3.6] = 6.4h$$

$$\Rightarrow h_2 = \frac{6.4}{1.2} h$$

$$P_0 + 0.8 \rho_w g h + \rho_w g (2h) = \rho_w g h_1 + P_0$$

$$\Rightarrow h_1 = h [0.8 + 2] \Rightarrow h_1 = 2.8 h$$

$$h_2/h_1 = 40/21$$

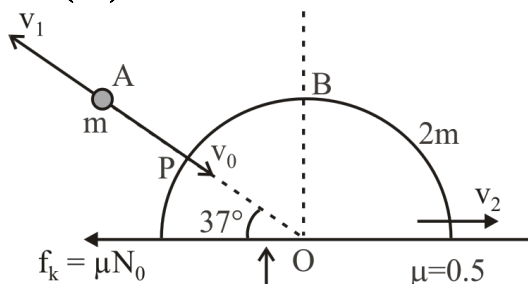
8. Ans (B)

$$P_0 + \frac{1}{2} \rho_w V^2 = P_0 + 0.8 \rho_w g h + \rho_w g (2h)$$

$$\Rightarrow \frac{v^2}{2} = gh [0.8 + 2]$$

$$v = \sqrt{2 \times 9.8 \times 0.28 \times 2.8} = 3.92 \text{ m/s}$$

10. Ans (D)



N_0 = normal force on the body B due to the horizontal surface

$$\int N_0 dt = \int N \sin 37^\circ dt$$

$$\int N_0 dt = \frac{3}{5} \int N dt \quad \dots (i)$$

$$\int f_k dt = \mu \int N_0 dt = 0.5 \times \frac{3}{5} \int N dt$$

$$\int f_k dt = 0.3 \int N dt \quad \dots (ii)$$

$$\int N dt = m(v_1 + v_0) \quad \dots (iii)$$

$$\int N \cos 37^\circ dt - \int f_k dt = 2mv_2$$

$$0.8 \int N dt - 0.3 \int N dt = 2mv_2$$

$$0.5 \int N dt = 2mv_2 \Rightarrow \int N dt = 4mv_2 \quad \dots (iv)$$

From equation (iii) and (iv)

$$m(v_1 + v_0) = 4mv_2$$

$$4v_2 - v_1 = v_0 \quad \dots (v)$$

$$0.6 = e = \frac{v_2 \cos 37^\circ + v_1}{v_0}$$

$$0.8v_2 + v_1 = 0.6v_0 \quad \dots (vi)$$

Solving equation (v) and (vi), we get

$$4.8v_2 = 1.6v_0$$

$$\Rightarrow v_2 = \frac{v_0}{3} = 5 \text{ m/s} \Rightarrow v_2 = 5 \text{ m/s}$$

From equation (v),

$$v_1 = 4v_2 - v_0 = 20 - 15 = 5 \text{ m/s}$$

$$v_1 = 5 \text{ m/s}$$

From equation (iii)

$$\int f_k dt = 0.3 \int N dt = 0.3 \times 20 = 6 \text{ N} - \text{S}$$

The impulse due to frictional force of the horizontal surface on the hemispherical body B during collision is

$$\int f_k dt = 6 \text{ N} - \text{S}$$

PART-1 : PHYSICS

SECTION-II (i)

2. Ans (3.00)

$$(3 - 0.5i)i = 4.32$$

$$0.5i^2 - 3i + 4.32 = 0$$

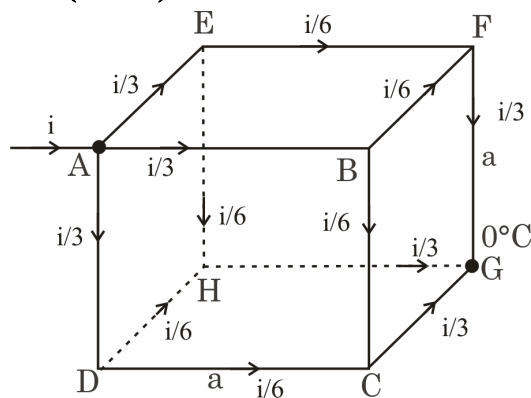
$$i = \frac{3 \pm \sqrt{9 - 28.64}}{1} = 2.4 \text{ A}$$

$$i = \frac{3}{0.5 + \frac{x}{1+x}}$$

$$\Rightarrow 0.5 + \frac{x}{1+x} = 1.25$$

$$x = 3\Omega$$

4. Ans (60.00)



$$\frac{i}{3} = \frac{(100 - T_E)}{R}$$

$$\frac{i}{6} = \frac{T_E - T_F}{R}$$

$$\frac{i}{3} = \frac{T_F - 0}{R}$$

$$100 = T_E + T_F$$

$$\frac{T_F}{2} = T_E - T_F$$

$$T_E = \frac{3T_F}{2}$$

$$\Rightarrow 100 = \frac{5T_F}{2} \Rightarrow T_F = 40$$

5. Ans (140.00)

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{yA}{m\ell}}$$

$$\omega = \sqrt{\frac{4 \times 10^9 \times 4.9 \times 10^{-7}}{0.1 \times 1}} = 140$$

6. Ans (0.50)

$$\frac{mg}{A} = \frac{Y \Delta \ell}{\ell_0}$$

$$x = \frac{mg\ell_0}{YA} = \frac{0.1 \times 9.8 \times 1}{4 \times 10^9 \times 4.9 \times 10^{-7}} = \frac{1}{2000}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (4)

$$R_1 = R_2 \Rightarrow \frac{mv}{qB} = \frac{T}{BI} \Rightarrow T = \frac{mvI}{Q}$$

$$\Rightarrow T = \frac{10^{-6} \times 10^6 \times 2}{500 \times 10^{-3}} = 4 \text{ newton}$$

8. Ans (9)

$$\text{Intensity at the surface of sphere } \rho 2\pi^2 f^2 (\Delta R)^2 \sqrt{\frac{B}{\rho}}$$

$$\text{So, total power} = \rho 2\pi^2 f^2 (\Delta R)^2 \rho \sqrt{\frac{B}{\rho}} 4\pi R^2$$

$$= 8\pi^3 \sqrt{\rho B} f^2 R^2 (\Delta R)^2 = 9 \text{ W att}$$

9. Ans (3)

$$3.95 = \frac{hc}{e\lambda} = \frac{\phi}{e} = 13.6Z^2 \left(\frac{1}{9} - \frac{1}{16} \right) - \frac{\phi}{e}$$

$$0.754 = 13.6Z^2 \left(\frac{1}{18} - \frac{1}{25} \right) - \frac{\phi}{e}$$

$$3.95 = \frac{13.6 \times 7}{144} Z^2 - \frac{\phi}{e}$$

$$0.754 = \frac{13.6 \times 9Z^2}{400} - \frac{\phi}{e}$$

$$3.95 - 0.754$$

$$= 13.6Z^2 \left[\frac{7}{144} - \frac{9}{400} \right]$$

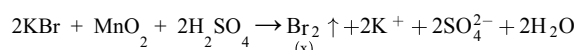
$$Z = \sqrt{\frac{3.196}{13.6} \times \frac{144 \times 400}{2800 - 9 \times 144}}$$

$$Z = \frac{240}{80} = 3$$

PART-2 : CHEMISTRY

SECTION-I (i)

5. Ans (A,C,D)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,C,D)

$$S = {}^nC_0^2 + \sum_{r=1}^n r \cdot \frac{{}^nC_r^2}{{}^nC_{r-1}}$$

$$S = 1 + \sum_{r=1}^n r \cdot {}^nC_r \cdot \frac{{}^nC_r}{{}^nC_{r-1}}$$

$$= 1 + \sum_{r=1}^n r \cdot {}^nC_r \cdot \frac{n-r+1}{r}$$

$$= 1 + \sum_{r=1}^n (n+1) \cdot {}^nC_r - \sum_{r=1}^n r \cdot {}^nC_r$$

$$= 1 + (n+1)2^n - n - 1 - n \cdot 2^{n-1}$$

$$= n \cdot 2 \cdot 2^{n-1} + 2^n - n - n \cdot 2^{n-1}$$

$$S = n \cdot 2^{n-1} + 2^{n-1} \cdot 2 - n$$

$$S + 50 = 50 \cdot 2^{49} + 2 \cdot 2^{49}$$

$$= 2^{49} \cdot 52$$

$$= 2^{49} \cdot 2 \cdot 26$$

$$= 2^{50} \cdot 26$$

2. Ans (A,C)

Consider 10, $h_2, h_3, h_4, \dots, h_{99}, 2020$ H.P

$$\frac{1}{10}, \frac{1}{h_2}, \frac{1}{h_3}, \frac{1}{h_4}, \dots, \frac{1}{h_{99}}, \frac{1}{2020} \rightarrow \text{A.P}$$

Multiply with 20200.

$$\frac{20200}{10}, \frac{20200}{h_2}, \frac{20200}{h_3}, \frac{20200}{h_4}, \dots, \frac{20200}{h_{99}} \rightarrow \text{A.P}$$

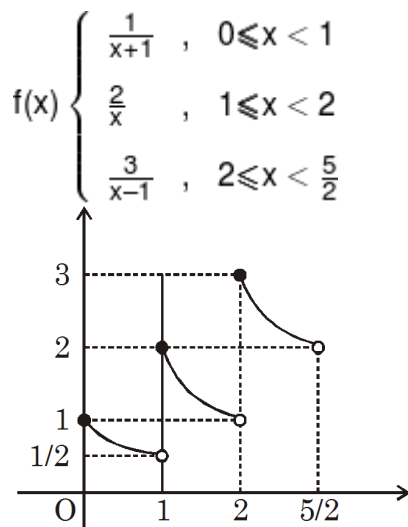
$$\frac{20200}{2020}, \frac{20200}{h_{99}}, \frac{20200}{h_{98}}, \dots, \frac{20200}{h_2}, \frac{20200}{10} \rightarrow \text{A.P}$$

We know 10, $a_2, a_3, a_4, \dots, a_{99}, 2020$ A.P.

$$\therefore a_2 = \frac{20200}{h_{99}}, a_3 = \frac{20200}{h_{98}}$$

$$a_{20} = \frac{20200}{h_{81}} \text{ and } a_{50} = \frac{20200}{h_{51}}$$

3. Ans (A,B,D)



Clearly $f(x)$ is discontinuous and bijective function

$$\lim_{x \rightarrow 1^-} f(x) = \frac{1}{2}$$

$$\lim_{x \rightarrow 1^+} f(x) = 2$$

$$\min \left(\lim_{x \rightarrow 1^-} f(x), \lim_{x \rightarrow 1^+} f(x) \right) = \frac{1}{2} \neq f(1)$$

$$\max(1, 2) = 2 = f(1)$$

4. Ans (A,B,C)

$$\text{Given } \frac{d}{dx} \left(y \frac{dy}{dx} \right) = x \Rightarrow y \frac{dy}{dx} = \frac{x^2}{2} + c$$

$$\Rightarrow \frac{d}{dx} \left(\frac{y^2}{2} \right) = \frac{x^2}{2} + c$$

$$\Rightarrow y^2 = \frac{x^3}{3} + \alpha x + \beta \quad \text{given } y(0) = 1, y(1) = 1$$

$$\Rightarrow \beta = 1, \alpha = -\frac{1}{3}$$

$$\therefore y^2 = f(x) = \frac{x^3 - x + 3}{3}, f'(x) = 3x^2 - 1 > 0$$

for $x > 1$

$$\therefore f(x) \uparrow$$

$$f\left(\frac{1}{\sqrt{3}}\right) f\left(-\frac{1}{\sqrt{3}}\right) > 0 \quad f(x) = 0 \text{ has only one}$$

real root

5. Ans (A,B)

$$\left| z - \frac{z_1 + 2z_2}{3} \right| = \frac{1}{3} |z_1 - z_2| \text{ and}$$

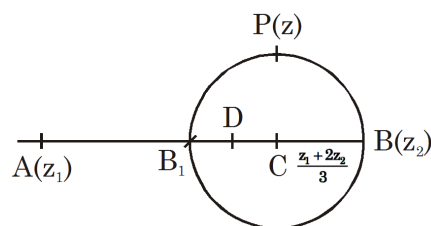
$$\text{Arg} \left(\frac{z_1 - z_2}{z - (kz_1 + (1-k)z_2)} \right) = \pm \frac{\pi}{2}$$

Angle between the line segment joining z_1 and z_2 ;

z and $kz_1 + (1-k)z_2$ is $\frac{\pi}{2}$ $\frac{z_1 + 2z_2}{3}$ is a point

on segment AB such that $AC : CB = 2 : 1$ and

$D(kz_1 + (1-k)z_2)$ is a point on segment AB such that $AD : DB = 1 - k : k \Rightarrow BD = k |(z_1 - z_2)|$



(a) for no sol

$$BD > BB_1 \Rightarrow k |z_1 - z_2| > \frac{2}{3} |z_1 - z_2| \Rightarrow k > \frac{2}{3}$$

(b) for more than one sol. $0 < BD < BB_1$

$$\Rightarrow 0 < k |z_1 - z_2| < \frac{2}{3} |z_1 - z_2|$$

$$0 < k < \frac{2}{3}$$

6. Ans (A,B,C,D)

$$\text{Slope of AS} = \frac{2at_1}{at_1^2 - a} = \frac{2t_1}{t_1^2 - 1}$$

$$\text{Slope of BC} = \frac{2a(t_3 - t_2)}{a(t_3^2 - t_2^2)} = \frac{2}{t_3 + t_2}$$

$$\therefore \frac{2t_1}{t_1^2 - 1} \times \frac{2}{t_3 + t_2} = -1$$

$$-4t_1 = t_3 t_1^2 + t_1^2 t_2 - t_3 - t_2$$

$$\therefore y t_1 = 0 \Rightarrow t_3 + t_2 = 0 \quad (C) \text{ is correct}$$

$$t_1^2(t_2 + t_3) + 4t_1 = t_3 + t_2$$

$$\text{paralarlly } t_2^2(t_1 + t_3) + 4t_2 = t_1 + t_3$$

$$t_2 t_1^2 + t_1^2 t_3 - t_2^2 t_1 - t_2^2 t_3 + 4(t_1 - t_2) = t_2 - t_1$$

$$t_1 t_2(t_1 - t_2) + t_3(t_1^2 - t_2^2) = +5(t_2 - t_1)$$

$$t_1 t_2 + t_3(t_1 + t_2) = -5$$

$$\therefore \sum t_1 t_2 = -5 \quad (A) \text{ is true}$$

$$\text{Now } t_1^2(t_2 + t_3) + 4t_1 = t_3 + t_2$$

$$t_1(t_1 t_2 + t_3 t_1) + 4t_1 = t_2 + t_3$$

$$t_1(-5 - t_2 t_3) + 4t_1 = t_2 + t_3$$

$$-t_1 t_2 t_3 - t_1 = t_2 + t_3$$

$$-t_1 t_2 t_3 = t_1 + t_2 + t_3$$

$$\therefore \frac{1}{t_1 t_2} + \frac{1}{t_1 t_3} + \frac{1}{t_3 t_1} = -1 \quad (B) \text{ is correct}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (C)**

Let $\left(2t, \frac{2}{t}\right)$ be a point on the hyperbola.

Equation of the tangent at this point $x + yt^2 = 8t$.

$A = (8t, 0)$, $B = (0, 8/t)$

Locus of circumcentre of triangle OAB is a hyperbola with its eccentricity is $= \sqrt{2}$

Shortest distance exist along the common normal.

$t^2 = \frac{1}{2} \Rightarrow t = \frac{1}{\sqrt{2}}$, foot of the perpendicular is

$(\sqrt{2}, 2\sqrt{2})$; shortest distance is $\frac{4(\sqrt{2}-1)}{\sqrt{5}}$. Let

$R(2,0)$ and $S(2 + r\cos\theta, r\sin\theta)$ lies on hyperbola

$|r_1 r_2| = \frac{8}{|\sin 2\theta|}$; minimum of $RS \times RT$ is 8

8. **Ans (D)**

Let $\left(2t, \frac{2}{t}\right)$ be a point on the hyperbola.

Equation of the tangent at this point $x + yt^2 = 8t$.

$A = (8t, 0)$, $B = (0, 8/t)$

Locus of circumcentre of triangle OAB is a hyperbola with its eccentricity is $= \sqrt{2}$

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$R(2,0)$ and $S(2 + r\cos\theta, r\sin\theta)$ lies on hyperbola

$|r_1 r_2| = \frac{8}{|\sin 2\theta|}$; minimum of $RS \times RT$ is 8

9. **Ans (C)**

$z^{10} - 1 = (z - 1)(z - \alpha)(z - \alpha^2) \dots (z - \alpha^9)$

where $\alpha = e^{i\left(\frac{2\pi}{10}\right)}$

Putting $z = 2i$ and taking modulus we get

$|(2i)^{10} - 1| = AB_0 \cdot AB_1 \cdot AB_2 \dots AB_9$

$\Rightarrow (AB_3 \cdot AB_4 \dots AB_7)^2 = |-2^{10} - 1|$ (by symmetry)

$\Rightarrow AB_3 \cdot AB_4 \dots AB_7 = \sqrt{1025} = 5\sqrt{41}$

10. **Ans (C)**

The product of distances

$$|\sqrt{2} - \sqrt{2}\alpha| |\sqrt{2} - \sqrt{2}\alpha^2| \dots |\sqrt{2} - \sqrt{2}\alpha^8|$$

$$= (\sqrt{2})^8 |z - \alpha| |z - \alpha^2| \dots |z - \alpha^8|, \text{ at } z = 1$$

$$= 16 \times \frac{|z^9 - 1|}{|z - 1|} = 16 \times |z^8 + z^7 + \dots + z + 1|$$

$$= 16 \times 9 = 144$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (10.00)**

$$I = 5 \int_0^{2\pi} \frac{\cos 6x \cos 7x \cos 8x \cos 9x dx}{1 + e^{2\sin^3 4x}}$$

$$= 5 \int_0^{2\pi} \frac{e^{2\sin^3 4x} \cos 6x \cos 7x \cos 8x \cos 9x dx}{1 + e^{2\sin^3 4x}}$$

$$2I = 5 \int_0^{2\pi} \cos 6x \cos 7x \cos 8x \cos 9x dx$$

$$= 20 \int_0^{\pi/2} \cos 6x \cos 7x \cos 8x \cos 9x dx$$

$$\therefore I = 10 \int_0^{\pi/2} \cos 6x \cos 7x \cos 8x \cos 9x dx$$

2. **Ans (5.00)**

$$I = 10 \int_0^{\pi/2} \cos 6x \cos 7x \cos 8x \cos 9x dx$$

$$= 10 \int_0^{\pi/2} \cos 6x \sin 7x \cos 8x \sin 9x dx$$

$$\therefore 2I = 10 \int_0^{\pi/2} \cos 6x \cos 8x \cos 2x dx$$

3. Ans (16.00)

Let $f(x) = ax^2 + bx + c$

Given $\int_a^b f(x)dx > 0$ for all $x > 0$

So $f(x) = 0$ has no real roots.

Now $g(0) = f''(0)f(0) = 2ac \Rightarrow ac = 6$

$\therefore$ a and c are the integers, so the possible values are

$a = 6, c = 1, a = 1, c = 6, a = 2, c = 3, a = 3, c = 2$

Again $b^2 < 4ac$ or $b^2 < 24$. So 'b' can be 1, 2, 3, 4

$\therefore$ in all $4 \times 4 = 16$ such quadratic are possible

now $f(1) = a + b + c$ then max. value of

$f(1) = 7 + 4 = 11$ and min $f(1) = 5 + 1 = 6$

4. Ans (6.00)

Let $f(x) = ax^2 + bx + c$

Given $\int_a^b f(x)dx > 0$ for all $x > 0$

So $f(x) = 0$ has no real roots.

Now $g(0) = f''(0)f(0) = 2ac \Rightarrow ac = 6$

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$a = 6, c = 1, a = 1, c = 6, a = 2, c = 3, a = 3, c = 2$

Again $b^2 < 4ac$ or $b^2 < 24$. So 'b' can be 1, 2, 3, 4

$\therefore$ in all $4 \times 4 = 16$ such quadratic are possible

now $f(1) = a + b + c$ then max. value of

$f(1) = 7 + 4 = 11$ and min $f(1) = 5 + 1 = 6$

5. Ans (6.00)

Obliviously $f(x)$ is a circulant matrix

$\therefore -f(x) = (\sin^2 x + \cos^2 x + \ell)$

$(\sin^4 x + \cos^4 x + \ell^2 - \sin^2 x \cos^2 x - \ell \cos^2 x - \ell \sin^2 x)$

$= (\ell + 1)(\ell^2 + 1 - 3\sin^2 x \cos^2 x - \ell)$

Now $f(x) = 0 \Rightarrow \ell^2 - \ell + 1 = 3\sin^2 x \cos^2 x = 0$

$\Rightarrow \ell^2 - \ell + 1 = \frac{3}{4} \sin^2 2x$

$16\ell^2 - 16\ell + 7 = 0 \Rightarrow 16\ell^2 - 16\ell + 16 = 9$

$\Rightarrow \sin^2 2x = \frac{3}{4} \Rightarrow \sin 2x = \frac{\sqrt{3}}{2}$ as we have to

find roots of $f(x) = 0$ in $\left(0, \frac{\pi}{2}\right)$

$\Rightarrow x = \frac{\pi}{6}, \frac{\pi}{3}$. Thus $f\left(\frac{\pi}{6}\right) = f\left(\frac{\pi}{3}\right) = 0$

further f is a continuous function and differentiable $\forall x \in \mathbb{R}$.

So finally $a = \frac{\pi}{6}, b = \frac{\pi}{3}$

6. Ans (3.00)

Obliviously $f(x)$ is a circulant matrix

$\therefore -f(x) = (\sin^2 x + \cos^2 x + \ell)$

$(\sin^4 x + \cos^4 x + \ell^2 - \sin^2 x \cos^2 x - \ell \cos^2 x - \ell \sin^2 x)$

$= (\ell + 1)(\ell^2 + 1 - 3\sin^2 x \cos^2 x - \ell)$

Now $f(x) = 0 \Rightarrow \ell^2 - \ell + 1 = 3\sin^2 x \cos^2 x = 0$

$\Rightarrow \ell^2 - \ell + 1 = \frac{3}{4} \sin^2 2x$

$16\ell^2 - 16\ell + 7 = 0 \Rightarrow 16\ell^2 - 16\ell + 16 = 9$

$\Rightarrow \sin^2 2x = \frac{3}{4} \Rightarrow \sin 2x = \frac{\sqrt{3}}{2}$ as we have to

find roots of $f(x) = 0$ in $\left(0, \frac{\pi}{2}\right)$

$\Rightarrow x = \frac{\pi}{6}, \frac{\pi}{3}$. Thus $f\left(\frac{\pi}{6}\right) = f\left(\frac{\pi}{3}\right) = 0$

further f is a continuous function and differentiable $\forall x \in \mathbb{R}$.

So finally $a = \frac{\pi}{6}, b = \frac{\pi}{3}$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (16)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(x) + f(h) + h\sqrt{f(x)} - f(x) - f(0) - 0\sqrt{f(x)}}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \left(\frac{f(h) - f(0)}{h - 0} \right) + \sqrt{f(x)}$$

$$f'(x) = \sqrt{f(x)}$$

$$\int \frac{f'(x)}{\sqrt{f(x)}} dx = \int dx$$

$$2\sqrt{f(x)} = x + c$$

$$f(x) = \frac{x^2}{4}$$

when $\alpha = 0$ area is minimum required minimum

area

$$= 2 \int_0^9 2\sqrt{y} dy \Rightarrow 4 \left(\frac{y^{3/2}}{3/2} \right)_0^9 = 72 \text{ sq. unit}$$

$$\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n \sqrt{r} \sum_{r=1}^n \frac{1}{\sqrt{r}}}{\frac{n(n+1)}{2}} \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n \sqrt{\frac{r}{n}} \sum_{r=1}^n \frac{1}{\sqrt{\frac{r}{n}}}}{\frac{n^2}{2} \left(1 + \frac{1}{n} \right)}$$

$$= 2 \times \int_0^1 \sqrt{x} dx \int_0^1 \frac{1}{\sqrt{x}} dx$$

$$= 2 \times \left[\frac{x^{3/2}}{3/2} \right]_0^1 \left[\frac{x^{1/2}}{1/2} \right]_0^1 = 2 \times \frac{2}{3} \times 2 = \frac{8}{3}$$

$$\therefore k = 8$$

$$\therefore A + K = 16$$

8. Ans (1)

$$\text{Let } x^4 + ax^3 + bx^2 + cx + d$$

$$= (x - \alpha)(x - \beta)(x - \gamma)(x - \delta)$$

$$\text{Let } (x - \alpha)(x - \beta) = x^2 + px + q$$

$$(x - \gamma)(x - \delta) = x^2 + tx + t$$

$$q = \alpha\beta \quad t = \gamma\delta$$

$$x^4 + ax^3 + bx^2 + cx + d$$

$$= x^4 + px^3 + (p^2 + q + t)x^2 + p(q + t)x + qt$$

$$a = 2p$$

$$b = p^2 + q + t \quad q + t = b - p^2 \dots(2)$$

$$c = pq + pt \Rightarrow q + t = \frac{c}{p} \dots(3)$$

$$d = qt \Rightarrow$$

$$(2) \text{ and } (3)$$

$$bp - p^3 = c$$

$$b \frac{a}{2} - \frac{a^3}{8} = c \quad [\text{from (1)}]$$

$$4ab - a^3 = 8c$$

$$a^3 - 4ab + 8c = 0$$

$$\text{Let } f(a) = a^3 - 4ab + 8c$$

$$f'(a) = 3a^2 - 4b > 0 \quad \{\because b < 0\}$$

$\therefore f(a)$ has exactly 1 real root

9. Ans (3)

$$2^7 \cos^3 \theta \sin^5 \theta$$

$$= (2 \sin \theta \cos \theta)^3 2^4 \sin^2 \theta$$

$$= 16 \sin^3 2\theta \sin^2 \theta$$

$$\text{We know } \sin 6\theta = 3 \sin 2\theta - 4 \sin^3 2\theta$$

$$= 4(3 \sin 2\theta - \sin 6\theta) \sin^2 \theta$$

$$= 2(3 \sin 2\theta - \sin 6\theta) (1 - \cos 2\theta)$$

$$= 2(3 \sin 2\theta - \sin 6\theta - 3 \sin 4\theta + \sin 6\theta \cos 2\theta)$$

$$= 6 \sin 2\theta - 2 \sin 6\theta - 3 \sin 4\theta + \sin 8\theta + \sin 4\theta$$

$$= 6 \sin 2\theta - 2 \sin 6\theta - 2 \sin 4\theta + \sin 8\theta$$

$$a = 1 \quad b = -2 \quad c = -2 \quad d = 6$$

$$\therefore a + b + c + d = 3$$